

Hand in

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Problem (1). Let $X = [-1, 1] \times \{-1, +1\} \subseteq \mathbb{R}^2$ have the usual metric topology and define a relation $\sim$ on X saying that $(a, -1) \sim (a, 1)$ for all $a \in [-1, 0) \cup (0, 1]$. Let $Y = X / \sim$ have the quotient topology

- Check that $\sim$ is an equivalence relation (nevermind I guess?)
- Is Y Hausdorff?
- Does Y have a countable basis for its topology?
- Check up the definition of connectedness in section 23: is Y connected?
- Check up the definition of compact in section 26: is Y compact?

Solution and proof of all.

Hausdorff?

Recall that Y is Hausdorff if any two distinct points have some open neighborhoods that are disjoint.

We will use the fact that open sets in X correspond exactly with open sets in Y .

For Y to have the Hausdorff property we would require that, as $(0, 1)$ and $(0, -1)$ are distinct, they have some open disjoint neighborhoods that are disjoint.

Denote $(a, 1) = a_+$ and $(0, -1) = a_-$. Let U be an open neighborhood of 0_+ and V an open neighborhood of 0_- .

Then $q^{-1}(U), q^{-1}(V)$ are open neighborhoods of X . As X has the metric topology we know there are $\varepsilon_+, \varepsilon_- > 0$ such that

$$(-\varepsilon_+, \varepsilon_+) \times \{1\} \subseteq q^{-1}(U)$$

and

$$(-\varepsilon_-, \varepsilon_-) \times \{-1\} \subseteq q^{-1}(V).$$

Choose $0 < a < \min\{\varepsilon_-, \varepsilon_+\}$. Then $a_+ \in q^{-1}(U)$ and $a_- \in q^{-1}(V)$, but

$$a_+ \sim a_-$$

so

$$q(a_+) = q(a_-) \in Y.$$

Call this point y . Then $y \in U \cap V$, meaning they are not disjoint. This violates the Hausdorff property. Intuitively, we will use the fact that some open neighborhood of $(0, 1)$ will have elements whose fibers

Countable basis?

Indeed it does have a countable basis.

X has a countable basis since it is just two copies of $[-1, 1] \subseteq \mathbb{R}$ which has a countable basis which we can construct using the rationals.

Let $\mathcal{B} = \{B_1, B_2, \dots\}$ be such a basis. Since q is open,

$$\mathcal{C} = \{q(B) \mid B \in \mathcal{B}\}$$

consists of open sets in Y . Moreover, if $y \in U$ for some open set in Y , choose $x \in q^{-1}(U)$. Since $q^{-1}(U)$ is open, there is some $B \in \mathcal{B}$ with

$$x \in B \subseteq q^{-1}(U).$$

Therefore $y \in q(B) \subseteq U$. So $\mathcal{C}$ is a countable basis for Y . □